Practical exercise on the grid training simulator

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Exercise 1

a) Determine from the screen “Zustandsübersicht”

- Current network loading : ..... MW
- Current network losses : .....MW
- Cost of losses per hour : .....Euro/h

b) Determine from the screen “Lastfluss 400 kV”:

- For the coal power plant in „Nordost 400“
  - Active power P: .....MW (+ is feeding)
  - Reactive power Q: ..... MVAr (+ is feeding inductively)
- For the line „West 400 - Nord 400“
  - Loading (%) : .....
  - Active power losses ΔP : .....MW
  - Reactive power generation/consumption ΔQ : .....MVAr

  (ΔQ is the sum of both reactive power measurements. $\Delta Q < 0$: consumption of inductive reactive power)

c) Determine from the screen “West 20”

- Active and reactive power and the current in the branch to the corridor “Festwiese”
  P = ...MW, Q= .....MVAr, Current I = ...A
- Busbar voltage for SS1 in West 20
  U_{1E}= U_{2E}= U_{3E}= U_{12}= in [kV]

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Exercise 2

Switching a line for circuit breaker maintenance:

- In Area „Mitte 20“ the circuit breaker of the branch to West 20 shall be maintained.
- Check if the switching bay is critically needed. If yes, switch to a alternative line, else switch off.
- Perform switching according to the 5 security rules! What is to be done centrally, what locally?
  - Switching: Open circuit breaker and open the disconnector on all sides.
  - Protect against reconnection: Set work markers („Arbeitsmarkierungen“) of type danger to all relevant switch components: „Revision CB by Müller till 2pm“
  - Check for zero voltage
  - Grounding
  - Covering of near devices under voltage
- Transfer disposition authorization (DA) from the control center to local work responsible (WR).
- The WR Reports the completion of the work.
- Take DA and restore the original state.

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Exercise 3

Bubar isolation in „UW Ost 20“

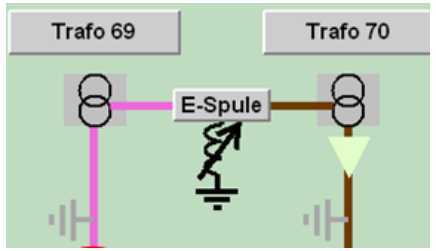
- In „UW OST 20“, the 20 kV busbar SS1 shall be isolated and replaced by SS2.
- The transformer feed and all junctions shall be transferred to SS2 without disruption. Strategy:
  - First close the busbar coupler
  - All branches of the „OST 20“ system must be switched from SS1 to SS2 "in sequence" without interruption.
  - The disconnector bridge always exists temporarily only
  - The transformer feed is switched, when about half of the load has been transferred.
  - Last, open the busbar coupler and the disconnectors, so the busbar is isolated.
- 2nd group member: Transfer all branches of substation OST-20 from SS-2 back to SS-1 without disruption
- 3rd group member:: reperform busbar isolation, so every one has done one iteration.

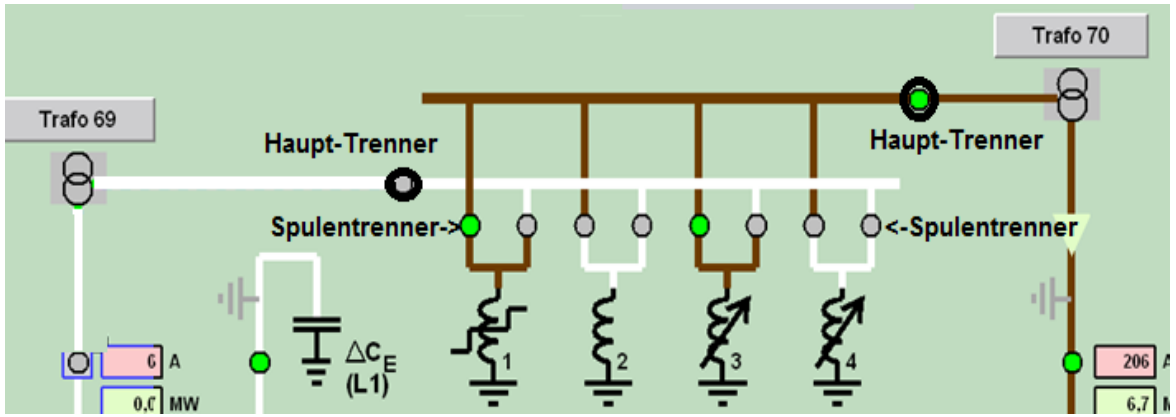
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Exercise 4

West 20 – Tranformer switch and authorization transfer

- In „West 20“, the conducting transformer shall be maintained by the grid service and shall be replaced disruption-free by the parallel transformer. The earthing coils shall be switched as well.
- Procedure (5 security rules shall be applied):
  - Set Voltage control of TR69 and TR70 to manual („Automatik AUS“)
  - Manually set voltages by increasing/decreasing the steps
  - Switch transformers to parallel operation (
  - Trafos parallel schalten (Condition: Voltage difference < 300V)
  - Switch earthing coils to the operating transformer
    - Select graphic earthing coil
    - Close the second busbar coil disconnector
    - Close the second starpoint main disconnector
    - Open the first starpoint main disconnector
    - Open the first busbar coil disconnector
  - Isolate , secure (marker „Gefahr“), ground TR 70
  - Transfer authorization to the Asset Management Service Team
- After the end of maintenance, reestablish authorization for the transformer
- Next group member, as above…





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Exercise 4b

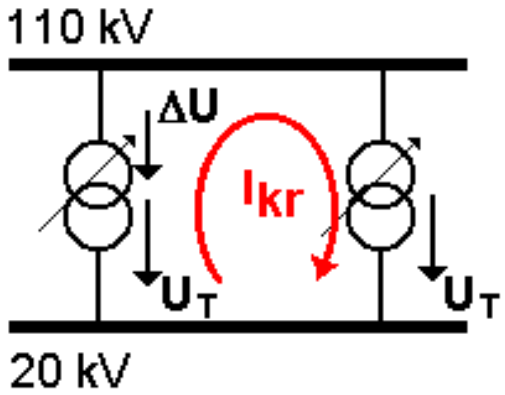
Parallel transformers and circulating currents

- The transformers TR 69 and 70 in „WEST 20“ are constructed identically.
- Turn both TR 69 and TR 70 in system „WEST 20“ to the same step and both voltage controllers to off („AUS“). Please note the load side values:

	Stufe	U [kV]	P [MW]	Q [MVAr]
Trafo 69				
Trafo 70				

- Change the step changer values by -4 for TR 69 and by +4 for TR 70

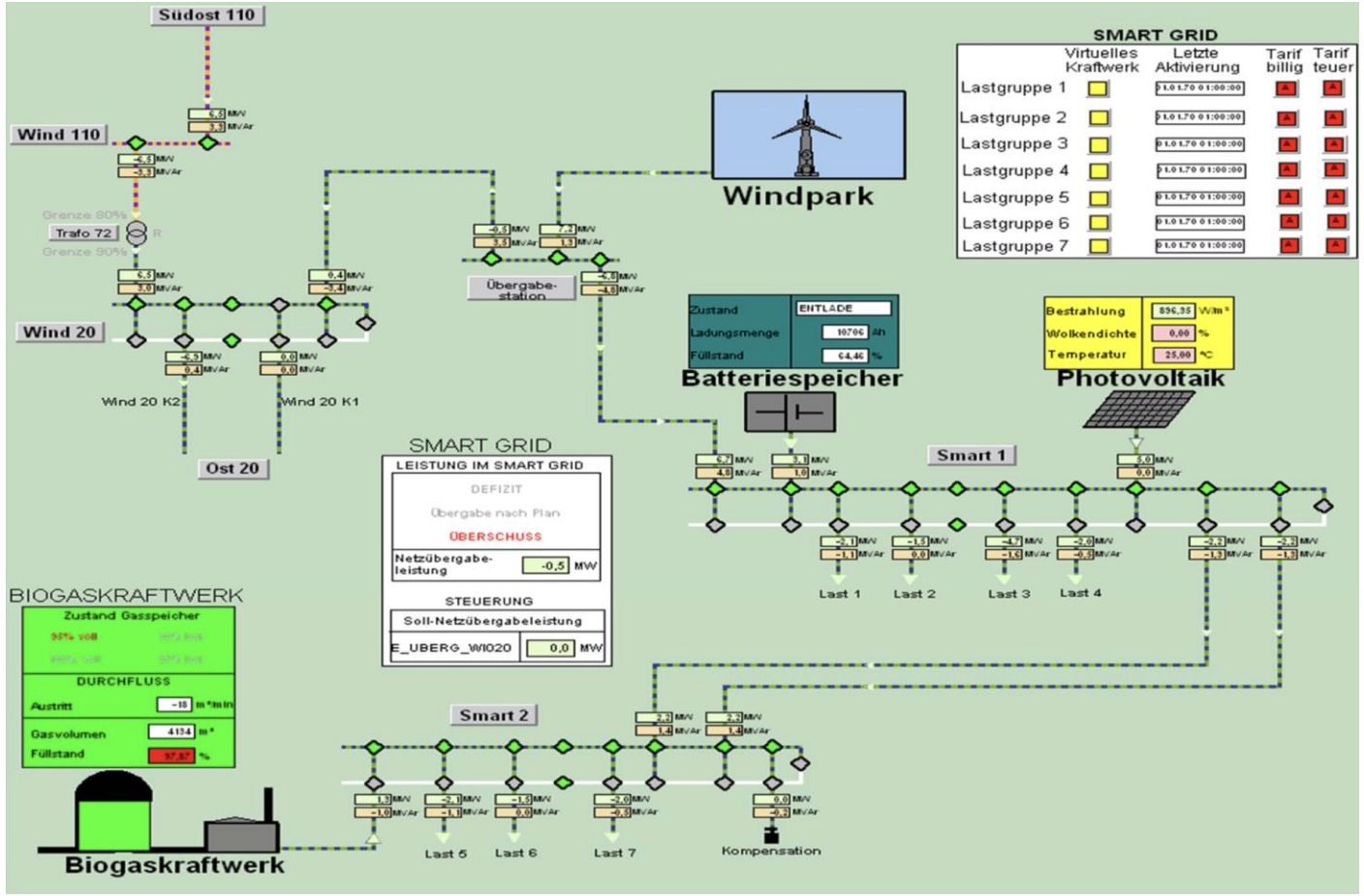
	Stufe	U [kV]	P [MW]	Q [MVAr]
Trafo 69				
Trafo 70				



- Explanation of the values: The set voltage difference (different numbers of steps) drives an additional (reactive) circulating current I_{Kr}

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Smart grid system



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Smart Grid Exercise: Overview I

- Select the „Smart Grid“ in the display chooser („Bildanwahl“, via the book icon at the top of the screen)
- In the following exercise you will analyse the behaviour of the load groups due to internal and external conditions
- Fist task: Identify the components of the smart grid:
  - Generators: ……
  - Loads: ……
- Determine the current active power values (right-click on measured value -> “Objekt Detail” -> displays values with more decimal places)
  - the bio gas plant……...[MW]
  - the wind park ……..[MW]
  - the photovoltaic plants ……..[MW]
  - the battery storage……..[MW] (+ “Entlade” = discharging/ - “Lade” = charging).
  - charging state …… [AH]

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Smart Grid Exercise: Components II

- Which of the load feeders has the highest power? ……..
- Determine the total load power: ……[MW]

……[MVAr]

- Which of the renewable injections has the most power in the current state?
  - ……..
  - Which value?……[MW]

Connection Point to the Smart Grid "Übergabestation"

- Is reactive power imported into the grid or exported?
  - ……
  - How much reactive power? ……[MVAr]
- Is reactive power imported into the grid or exported?
  - ……
  - How much reactive power? ……[MW]

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Smart Grid Exercise: Battery Storage I

- charging: $P_{DC} = \eta_{charge} \cdot P_{AC} \Rightarrow \Delta W_{DC} = \eta_{charge} \cdot \Delta W_{AC}$
- discharging: $P_{AC} = \eta_{discharge} \cdot P_{DC} \Rightarrow \Delta W_{AC} = \eta_{discharge} \cdot \Delta W_{DC}$
- charge / capacity Q in Ah: $\Delta Q = \int I(t)\, dt$
- $\Delta W_{DC} = U_{DC} \cdot \Delta Q$ (U_{DC}: Leerlaufspannung)
- energy / charging-/discharging power: $\Delta W_{AC} = \int P_{AC}(t)\, dt = P_{AC,mittel} \cdot \Delta t$
- Gesamtwirkungsgrad: $\eta_{totoal} = \eta_{charge} \cdot \eta_{discharge}$
  - special, if $\eta_{charge} = \eta_{discharge} = \eta \Rightarrow \eta = \sqrt{\eta_{total}}$
- Deep discharge protection: $Q > 20\% \cdot Q_{max}$

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Smart Grid Exercise: Battery Storage II

Determine the efficiency and the open circuit voltage:

- Open a “Kurvenfenster” and plot the values P_{AC}(MW) and Q(Ah)
- discharge the battery for 6 min with a power of 5 MW

(Open „Batteriespeicher“ -> double click on „P-Soll“ and adjust „Neuer Wert“)

  - determine $\Delta Q_{discharge} = Q(t_2) - Q(t_1) =$ …… (read ΔQ form the ruler in the curve window)
  - determine $\Delta W_{AC1} = P_{AC1} \cdot \Delta t_1 =$ ……

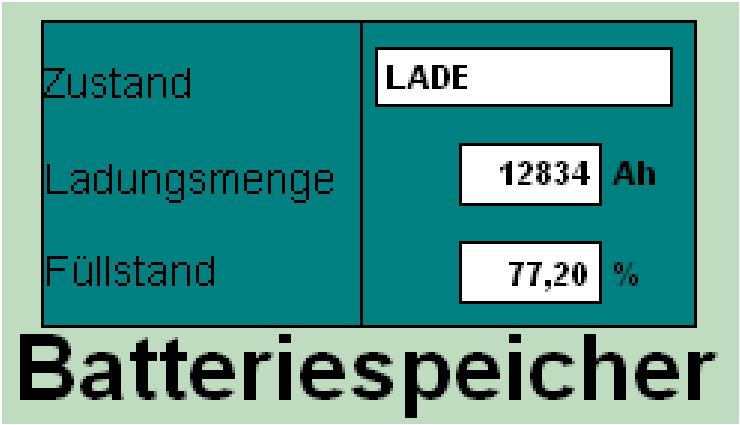
- charge the battery for 6 min at a power of (-) 5 MW
  - determine $\Delta Q_{charge} = Q(t_4) - Q(t_3) =$ ……
  - determine $\Delta W_{AC2} = P_{AC2} \cdot \Delta t_2 =$ ……

- determine $\eta_{total} = (\Delta W_{AC\ discharge} / \Delta Q_{discharge}) \cdot (\Delta Q_{laden} / \Delta W_{AC\ laden})$
- determine $\eta = \eta_{charge} = \eta_{discharge} = \sqrt{\eta_{total}}$ (assumption: identical efficiency)
- determine U_{DC} from $\Delta W_{DC} = U_{DC} \cdot \Delta Q$

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Smart Grid Exercise: Battery Storage III

- Question: Which time T_v can the battery supply 5 MW to the smart grid starting from 100% SOC. The battery should not be discharged by more than 80% (deep discharge protection)?
- T_V = ...

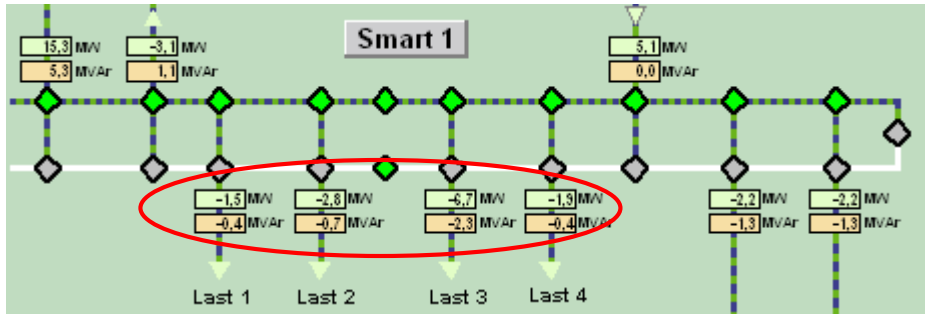


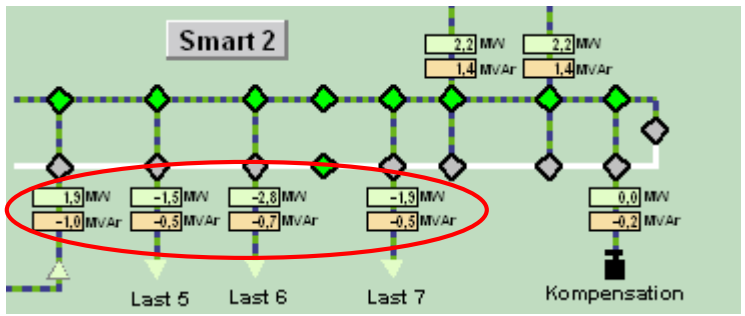
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Smart Grid Exercise: Battery Storage IV

- Determine the current total load in the smart grid [MW], the voltage of the battery [kV] and the current charging state of the battery [Ah]
  - ……MW
  - ……kV
  - ……Ah
- Which time T could the battery theoretically (without considering the 5MW limit) supply the current load of the smart grid (100% discharging)?





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Smart Grid Exercise: Battery Storage as a Controller

- Consider possible reasons for the installation of a battery in a today's network and justify your points (give 3 reasons)

- The battery storage was implemented to control the active power transfer to the feeder system ("Netzübergabeleistung"). This shall be analysed in detail:
  - Settings: Fahrplan [A], Batt. Regelung [E], Ver-Überg. [E]
  - Adjust the set point of the power transfer ("Soll-Netzübergabeleistung") so that the battery has a power of ≈0 MW.
  - Open a curve window and plot the actual and set point value of the transfer power as well as the battery power.
  - Change the set point value of the transfer power by ΔP = +3MW and observe the measured values in the curve window.
  - Now change the value of the transfer power by ΔP = -6MW and observe the measured values in the curve window.
  - Change the set point of the transfer power in steps of ΔP = -1MW and determine the power limit of the battery.

